Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2022-P1-Model)** **Date: 12-04-2026**

Time: 09.00Am to 12.00Pm **GTA-23** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**8**	2	**6**	3	**1**	4	**0.5**	5	**2.5**	6	**4**
7	**43**	8	**2**	9	**ABD**	10	**ABCD**	11	**AB**	12	**AC**
13	**AD**	14	**ABD**	15	**A**	16	**B**	17	**B**	18	**D**

PHYSICS

19	**4**	20	**1**	21	**1.15 - 1.25**	22	**4**	23	**1**	24	**6**
25	**375**	26	**2**	27	**CD**	28	**ABC**	29	**BCD**	30	**ABC**
31	**ABC**	32	**ABD**	33	**A**	34	**C**	35	**D**	36	**D**

CHEMISTRY

37	**88.89**	38	**6.25**	39	**0.56 – 0.58**	40	**39**	41	**17**	42	**1**
43	**0.65**	44	**10.25**	45	**ABCD**	46	**ABC**	47	**BCD**	48	**ACD**
49	**ACD**	50	**ABD**	51	**D**	52	**C**	53	**C**	54	**C**

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SOLUTIONS
MATHEMATICS

1. M = 0, m = 8
2. Roots are $-1,2,3,\&7$
3. $(1,1),(1,2),(2,1),(2,2),(2,3),(3,1),(3,2),(3,3)$
4.
5. $t_n=\dfrac{n+2}{n(n+1)(n+3)}$

$=\dfrac{1}{6}\left[\dfrac{1}{n+1}-\dfrac{1}{n+2}\right]+\dfrac{1}{6}\left[\dfrac{1}{n+2}-\dfrac{1}{n+3}\right]+\dfrac{2}{3}\left[\dfrac{1}{n}-\dfrac{1}{n+1}\right]$

So, $S=\sum_{n=1}^{n}T_n=\dfrac{1}{6}\sum_{n=1}^{n}\left(\dfrac{1}{n+1}-\dfrac{1}{n+2}\right)+\dfrac{1}{6}\sum_{n=1}^{n}\left(\dfrac{1}{n+2}-\dfrac{1}{n+3}\right)+\dfrac{2}{3}\sum_{n=1}^{n}\left(\dfrac{1}{n}-\dfrac{1}{n+1}\right).$

$=\dfrac{1}{6}\left(\dfrac{1}{2}-\dfrac{1}{n+1}\right)+\dfrac{1}{6}\left(\dfrac{1}{3}-\dfrac{1}{n+1}\right)+\dfrac{2}{3}\left(1-\dfrac{1}{n+1}\right)$

$=\dfrac{1}{6}\left[\dfrac{29}{6}-\dfrac{4}{n+1}-\dfrac{1}{n+2}-\dfrac{1}{n+3}\right](proved)$

6. m = 2 (-2 &4, 2,10 are not in the domain)
7. A=5, b =8, c = 9 & d = 11
8. STATEMENT 1 :

Let $\overrightarrow{OA}=\hat{a},\overrightarrow{OB}=\hat{b},\overrightarrow{OC}=\hat{c}$ and $\overrightarrow{OP}=\hat{d}$

Here, $\hat{a}\cdot\hat{b}=\hat{b}\cdot\hat{c}=\hat{c}\cdot\hat{a}=\dfrac{1}{2}$

And $\hat{d}.\hat{a}=\hat{d}.\hat{b}=\hat{d}.\hat{c}=\cos\alpha$

Let, $\hat{d}=x\hat{a}+y\hat{b}+z\hat{c}$

$\hat{d}\cdot\hat{a}=x+\dfrac{y}{2}+\dfrac{z}{2}=\cos\alpha\Rightarrow\hat{d}\cdot\hat{b}=\dfrac{x}{2}+y+\dfrac{z}{2}=\cos\alpha\Rightarrow\hat{d}\cdot\hat{c}=\dfrac{x}{2}+\dfrac{y}{2}+z=\cos\alpha$

Adding we get $2(x+y+z)=3\cos\alpha$

Taking dot product with $\hat{d}$ in (1) we get

$1=x\cos\alpha+y\cos\alpha+z\cos\alpha\Rightarrow x+y+z=\dfrac{1}{\cos\alpha}$

$\Rightarrow\dfrac{3\cos\alpha}{2}=\dfrac{1}{\cos\alpha}\Rightarrow\cos^2\alpha=\dfrac{2}{3}\Rightarrow 3p_1=3\cos^2\alpha=2$

STATEMENT 2 :

Note that $\overrightarrow{AB}^2+\overrightarrow{CD}^2=3^2+11^2=130=7^2+9^2=\overrightarrow{BC}^2+\overrightarrow{DA}^2$

Since $\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CD}+\overrightarrow{DA}=\vec{0}$,

$DA^2=\left(\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CD}\right)^2=AB^2+BC^2+CD^2+2\left(\overrightarrow{AB}.\overrightarrow{BC}+\overrightarrow{BC}.\overrightarrow{CD}+\overrightarrow{CD}.\overrightarrow{AB}\right)$

$=AB^2-BC^2+CD^2+2\left(\overrightarrow{BC}^2+\overrightarrow{AB}.\overrightarrow{BC}+\overrightarrow{BC}.\overrightarrow{CD}+\overrightarrow{CD}.\overrightarrow{AB}\right)$

$=AB^2-BC^2+CD^2+2\left(\overrightarrow{AB}+\overrightarrow{BC}\right).\left(\overrightarrow{BC}+\overrightarrow{CD}\right),$

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$i.e., 2\overrightarrow{AC}.\overrightarrow{BD} = 2\left(\overrightarrow{AB}+\overrightarrow{BC}\right).\left(\overrightarrow{BC}+\overrightarrow{CD}\right) = AD^2 + BC^2 - AB^2 - CD^2 = 0$

Hence both AC and BD are perpendicular to each other

$\Rightarrow p_2 = \cos\theta = 0$

9. $f(x) = x^2 + 5x - 4$

10. $f(x) = \frac{x}{2}\left(e^{\frac{x^2}{2}+\frac{x}{6}}\right)$

11. A) $x_n = \frac{1}{2}\left(x_{n-1} + \frac{5}{x_{n-1}}\right) = \frac{\sqrt{5}}{2}\left(\frac{x_{n-1}}{\sqrt{5}} + \frac{\sqrt{5}}{x_{n-1}}\right)$

$\Rightarrow \frac{x_n}{\sqrt{5}} = \frac{1}{2}\left(\frac{x_{n-1}}{\sqrt{5}} + \frac{5}{x_{n-1}}\right).$

By componendo-dividend, we get

$$\Rightarrow \frac{x_n - \sqrt{5}}{x_n + \sqrt{5}} = \frac{\frac{1}{2}\left(\frac{x_{n-1}}{\sqrt{5}} + \frac{\sqrt{5}}{x_{n-1}}\right) - 1}{\frac{1}{2}\left(\frac{x_{n-1}}{\sqrt{5}} + \sqrt{\frac{5}{x_{n-1}}}\right) + 1} = \frac{x_{n-1}^2 + \sqrt{5} - 2\sqrt{5}x_{n-1}}{x_{n-1}^2 + \sqrt{5} + 2\sqrt{5}x_{n-1}}$$

$$\Rightarrow \left(\frac{x_{n-1} - \sqrt{5}}{x_{n-1} + \sqrt{5}}\right)^2 = \left(\frac{x_{n-2} - \sqrt{5}}{x_{n-2} + \sqrt{5}}\right)^{2^2} = \ldots\ldots = \left(\frac{x_1 - \sqrt{5}}{x_1 + \sqrt{5}}\right)^{2n-1}$$

$$= \left(\frac{x_1 - \sqrt{5}}{x_1 + \sqrt{5}}\right)^{2n} \text{ (proved)}$$

B) Since x is a positive number, $x - \sqrt{5} < x + \sqrt{5}$

$$\Rightarrow \frac{x - \sqrt{5}}{x + \sqrt{5}} < 1$$

$$\therefore \left(\frac{x - \sqrt{5}}{x + \sqrt{5}}\right)^{2n} \to 0 \text{ as } n \to \infty.$$

$$\Rightarrow \frac{x - \sqrt{5}}{x + \sqrt{5}} \to 0 \Rightarrow x - \sqrt{5} \to 0$$

$$\therefore \lim_{n\to\infty} x_n = \sqrt{5}.$$

12.

13. $P = \begin{pmatrix}1&0&0\\0&1&1\\0&1&0\end{pmatrix}, \begin{pmatrix}1&0&1\\0&1&0\\1&0&0\end{pmatrix}, \begin{pmatrix}1&0&0\\0&0&1\\0&1&1\end{pmatrix}, \begin{pmatrix}1&1&0\\1&0&0\\0&0&1\end{pmatrix}, \begin{pmatrix}0&1&0\\1&1&0\\0&0&1\end{pmatrix}, \begin{pmatrix}0&0&1\\0&1&0\\1&0&1\end{pmatrix},$

14. Clearly, $\vec{a} = AB + AC + AD$

Also it is given that, $\overrightarrow{AB} \times \overrightarrow{AC} = \vec{b}$ and $\overrightarrow{AB} \times \overrightarrow{AD} = \vec{c}$

$\Rightarrow (\overrightarrow{AB} \times \overrightarrow{AC}) \times \vec{a} = \vec{b} \times \vec{a}$ and $(\overrightarrow{AB} \times \overrightarrow{AD}) \times \vec{a} = \vec{c} \times \vec{a}$

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Also given $\frac{\overrightarrow{AB}\cdot\vec{a}}{|\vec{a}|}=1,\frac{\overrightarrow{AC}\cdot\vec{a}}{|\vec{a}|}=1\Rightarrow\overrightarrow{AB}\cdot\vec{a}=3,\overrightarrow{AC}\cdot\vec{a}=3$

Solving (1) and (2) and (3) we get desired result

15. $f(x)=0$

$g(x)=\pi$

$h(x)=\begin{cases}0,x\in[0,1)\\ \pi,x\in[1,\infty)\end{cases}$

16. a) $\left\{\frac{3\pi}{10},n\in z\right\}$

b) $\left\{\frac{3\pi}{3}+\frac{\pi}{6},\frac{2\pi}{5}+\frac{\pi}{10},n\in z\right\}$

c) $\left\{\pi n+\frac{n\pi}{5}+\frac{\pi}{10},n\in z\right\}$

d) $\left\{\pi n+\frac{n\pi}{3}+\frac{\pi}{6},n\in z\right\}$

17. $\left[\begin{array}{l}\log_{10}(2xy)=4+(\log_{10}x-1)(\log_{10}y-2)\\ \log_{10}(2yz)=4+(\log_{10}y-2)(\log_{10}z-1)\\ \log_{10}(zx)=2+(\log_{10}z-1)(\log_{10}x-1)\end{array}\right\}$

Let $\log x-1=A$

$\log y-2=B$

$\log_{10}z-1=C$

In 'S'

$\begin{cases}4+AB=\log 2+A+1+B+2\Rightarrow(A-1)(B-1)=\log 2\\ 4+BC=\log 2+B+2+C+1\Rightarrow(B-1)(C-1)=\log 2\\ 2+AC=\quad C+1+A+1\Rightarrow(A-1)(C-1)=1\end{cases}$

$\Rightarrow(A-1)(B-1)(C-1)=\pm\log 2$

$C-1=\pm1$

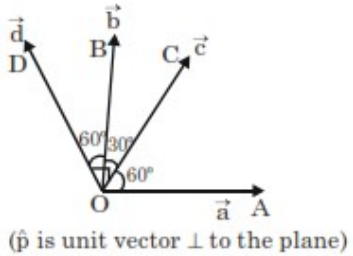
$A-1=\pm1$

$B-1=\pm\log 2$

$(A,B,C)\equiv(2,1+\log 2,2),(0,1-\log 2,0)$

$(x,y,z)\equiv(10^3,2000,10^3),(10,500,10)$

18.



($\hat{p}$ is unit vector $\perp$ to the plane)

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$\sin\theta_1 \sin\theta_2 \cos\phi = 1, \cos\alpha = \frac{1}{2}$

$\theta_1 = \theta_2 = 90°, \phi = 0, \alpha = \frac{\pi}{3}$

$\Rightarrow$OABCD are coplanar

$\Delta ABC = \frac{1}{2}\left|\vec{a}\times\vec{b} + \vec{b}\times\vec{c} + \vec{c}\times\vec{a}\right|$

$= \frac{1}{2}\left|2\hat{p} - 3\hat{p} - \frac{3\sqrt{3}}{2}\hat{p}\right| = \frac{1}{2}\left(1 + \frac{3\sqrt{3}}{2}\right)$

$\Delta BCD = \frac{1}{2}\left|\vec{b}\times\vec{c} + \vec{c}\times\vec{d} + \vec{d}\times\vec{b}\right|$

$= \frac{1}{2}\left|-3\hat{p} + 12\hat{p} - 4\sqrt{3}\hat{p}\right| = \frac{1}{2}\left(9 - 4\sqrt{3}\right)$

PHYSICS

19. $l = 94 \pm 0.1 cm$

$r = 5 \pm 0.05 cm$

$f = \frac{v}{4L}, L = l + 1.2r$

$\frac{\Delta f}{f}\times 100 = \frac{\Delta L}{L}\times 100$

$\alpha^2 = \frac{\Delta l + 1.2\times\Delta r}{l + 1.2\times r}\times 100$

$= 0.16$

$\alpha = 0.4, 10\alpha = 4$

20. $mg\cos\theta + kx = \frac{mv^2}{R}$ ____(1)

$mg\frac{R}{2} = \frac{1}{2}mv^2$ ____(2)

$mv^2 = mg$

$kx = \frac{mg}{2}$

$x = \frac{2\times 10}{2\times 20} = 0.5$

$l_0 + x = R$

$l_0 = 1m$

21. $V_0 = \sqrt{\frac{GM}{R}}$

The first impulse point is apogee (R) of the elliptical path perigee must be 2R.

Velocity of apogee = $v_0 + \Delta v_1$

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Velocity of perigee $= \dfrac{v_0 + \Delta v_1}{2}$

$$\frac{v_0 + \Delta v_1}{2} + \Delta v_2 = \sqrt{2}\, v_0$$

Also, $\dfrac{-GMm}{3R} = \dfrac{-GMm}{R} + \dfrac{1}{2} m\left(v_0 + \Delta v_1\right)^2$

$$\frac{2GMm}{3R} = \frac{1}{2} m\left(v_0 + \Delta v_1\right)^2$$

$$\sqrt{\frac{4GM}{3R}} = \left(v_0 + \Delta v_1\right)$$

$$\frac{2}{\sqrt{3}} v_0 = v_0 + \Delta v_1$$

$$\Delta v_1 = \left(\frac{2}{\sqrt{3}} - 1\right) v_0$$

$$\Delta v_2 = \frac{1}{\sqrt{2}} v_0 - \frac{1}{\sqrt{3}} v_0 \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{3}}\right) v_0$$

22. $\left[L^3 T^{-1}\right] = \left[ML^{-1}T^{-1}\right]^a [L]^b \left[ML^{-2}T^{-2}\right]^c$

$\Rightarrow a + c = 0$

$-a + b - 2c = 3$

$\Rightarrow a + b = 3$

and $-a - 2c = -1$

$\Rightarrow c = 1, a = -1, b = 4$

$a + b + c = 4$

23. $V_0 = 1 cm^3$

$\gamma = 1.8 \times 10^{-4} K^{-1}$

$A = 1.8 \times 10^{-4} cm^2$

$\Delta V = V_0 \gamma \Delta T = A \times \Delta h; \Delta T = 1K$

$$\Delta h = \frac{1 \times 1.8 \times 10^{-4}}{1.8 \times 10^{-4}} cm = 1 cm$$

24. $x_1 + x_2 = x; x = \dfrac{27}{4} mm, x_1 = ?$

$$F = \frac{\pi r^2 Y}{2L} x_1 = \frac{4\pi r^2 Y}{L} x_2$$

$$x_2 = \frac{x_1}{8}$$

$$\frac{9x_1}{8} = \frac{27}{4} \Rightarrow x_1 = 6 mm$$

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25. $\delta = 10^4\, kg/m^3$

$\sigma = \delta \times 10^8\, N/m^2$

$m = 4\pi\, gm$

$T = 900\pi N$

$f = \frac{v}{2L} = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$

$= \frac{T\delta}{2m\sigma}\sqrt{\frac{\sigma}{\delta}}$

$= \frac{T\delta}{2m}\sqrt{\frac{\sigma}{\delta}}$

$= \frac{900\pi}{2\times 4\pi \times 10^{-3}}\sqrt{\frac{10^4}{9\times 10^8}}$

$\frac{9000}{8\times 3} = \frac{3000}{3} = 375\, Hz$

$\frac{T}{A} = \sigma ___(1)$

$\mu = \frac{m}{L} _____(2)$

$\frac{\mu}{A} = \delta ___(3)$

from (1) & (3)

$\frac{T}{\mu} = \frac{\sigma}{\delta}$

$\frac{1}{L} = \frac{T\delta}{m\sigma}$

26. $V = \frac{dx}{dt}; a = V\frac{dv}{dx}$

$E = \frac{1}{2}mv^2 + 2\times\frac{1}{2}\times\frac{3mR^2}{2}\times\frac{V^2}{R^2} + \frac{1}{2}Kx^2$

$= 2mV^2 + \frac{1}{2}Kx^2$

$\frac{dE}{dx} = 0 \Rightarrow 4ma + Kx = 0$

$a = -\left(\frac{K}{4m}\right)x$

$T = 4\pi\sqrt{\frac{m}{K}}; \beta = 2$

27. Applying work energy theorem $W_{gas} + W_{atm} = \Delta KE = -\frac{1}{4}mu^2$

$\Rightarrow W_{gas} = -\frac{1}{4}mu^2 - W_{atm} = -\Delta U_{gas}$

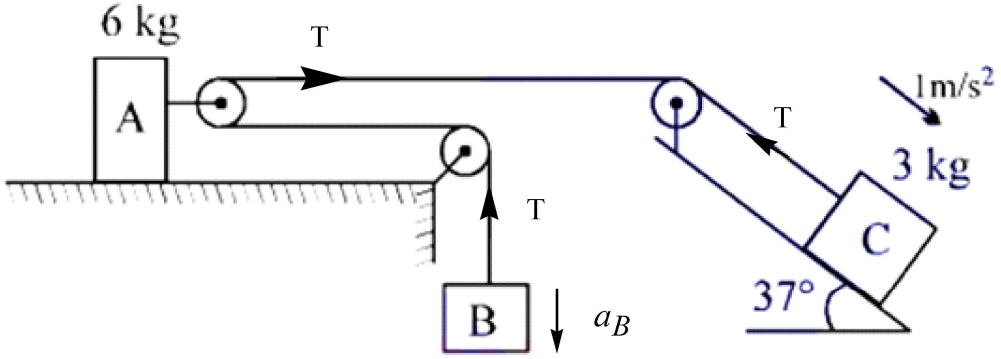
Since the gas is compressed

$\therefore W_{gas} > 0$

$\Rightarrow W_{gas} < -\frac{1}{4}mu^2 \; and \; \Delta U_{gas} > \frac{1}{4}mu^2$

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$3\times10\times\frac{3}{5}-T=3\times1$

$T=15N$

$mg-T=m\times g$

$m=15kg$

$G_A=5m/s^2$

$a_B=9m/s^2,\downarrow$

29. $F_1-F_2=(m_1+m_2)a_0$

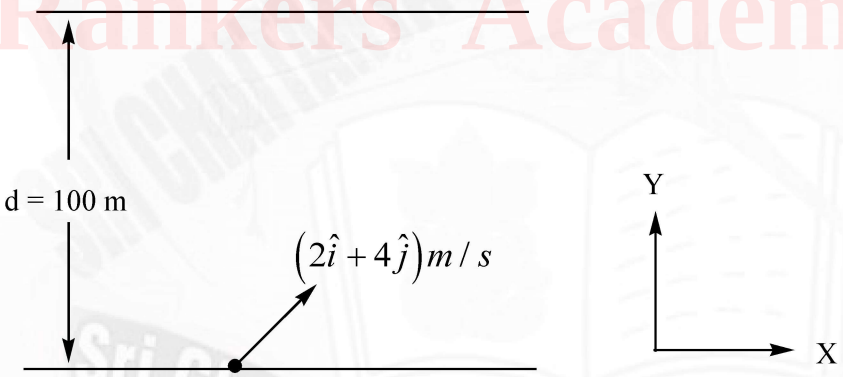
$kx-F_2=m_2a_0$

$kx=m_2a_0+F_2$

$a_2=\frac{kx}{m_2}=a_0+\frac{F_2}{m_2}$

$a_1=a_0$

30.



$\vec{u}_b=\vec{u}_{b\to r}+\vec{u}_{r\to e}$

$u_b=2\sqrt{5}m/s$

$=-2\hat{i}+4\hat{j}+4\hat{i}$

$=(2\hat{i}+4\hat{j})m/s$

$d=100m$

$T=\frac{100}{4}=25s$

$drift=25\times2=50m$

31. $\mu mg=10N$

$20N=m\times10m/s^2$

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$m = 2kg, \mu = 0.5$

32. Beat frequency = $|f_1 - f_2|$.

33. $m = 2kg, R = 11N/m$

A: $R_{eq} = 200\,N/m$ $\qquad a = 2m/s^2$

B: $a = \dfrac{1(3+1)}{2} = 2m/s^2$

C: $a = \dfrac{1}{2}(6+2) = 4m/s^2$

D: $a = \dfrac{1}{2}(5-3) = m/s^2$

34. Kinetic energy in com frame is always zero.

35. A: $A: I = \dfrac{Ml^2}{3}\sin^2\theta = \dfrac{MR^2}{12}$

B: $MR^2 = I_{cm} + M\left(\dfrac{2R}{\pi}\right)^2$ -----(1)

$I = I_{cm} + M\left(R - \dfrac{2R}{\pi}\right)^2$ -------(2)

$I = MR^2 + MR^2\left\{\left(1-\dfrac{2}{\pi}\right)^2 - \left(\dfrac{2}{\pi}\right)^2\right\}$

$= MR^2\left[1+1-\dfrac{4}{\pi}\right]$

$= 2MR^2\left(1-\dfrac{7}{\pi}\right) = \dfrac{SMR^2}{11}$

C: $I = \dfrac{1}{6}M\left(\dfrac{R\sqrt{3}}{2}\right)^2 = \dfrac{MR^2}{8}$

D: $i = \dfrac{M(2R)^2}{2} - \dfrac{M}{4}\left\{\dfrac{R^2}{2} + R^2\right\} = MR^2\left[2 - \dfrac{3}{8}\right]$

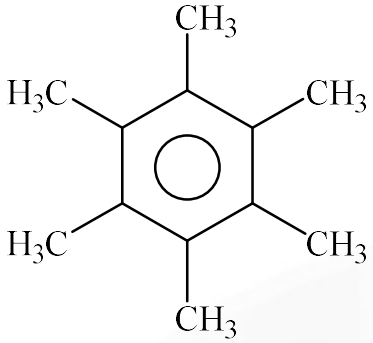
$= \dfrac{13MR^2}{8}$

36. For siphon to work: $h_1 + h_2 \le \dfrac{P}{\rho g}$

CHEMISTRY

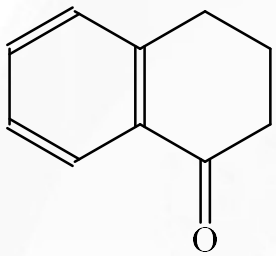
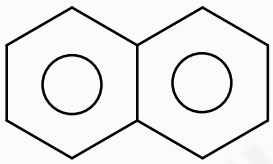
37. $P: CH_3 - C \equiv C - CH_3$

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Q:  $(C_{12}H_{18})$

$$w\%\text{ of } C = \frac{144}{162}\times 100 = 88.89\%$$

38.

X is  , Y is  $C_{10}H_8$

39. $Vlt\,1MS + Vlt\,1M\,(S,R)$

$$V\,\text{mol}\,S + \frac{V}{2}\text{mol}\,S + \frac{V}{2}\text{mol}\,R$$

$$[\alpha] = \frac{3V}{2\times 2V}\times(+13) + \frac{V}{2\times 2V}\times(-13) = +6.5^\circ$$

$$+6.5 = \frac{\alpha\times 1000}{1\times 88} \Rightarrow \alpha = 6.5\times 88/1000 = 0.57$$

40. $X = 120\left(sp^2 - \text{hybridization}\right)$

$Y = 180(sp - \text{hybridization})$

$Z = 90\left(sp^3d^2 - \text{hybridization}\right)$

$$\frac{120+180+90}{10} = \frac{390}{10} = 39$$

41. In trigonal bipyramidal angels are 9 and square pyramidal angels are 8

42.

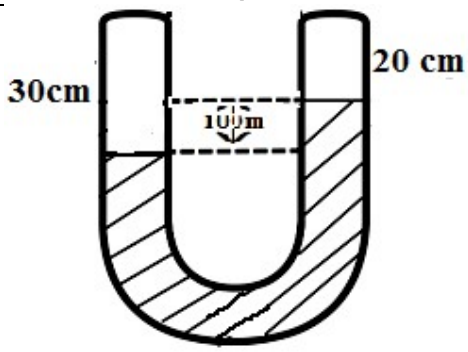
	$NH_3(g)$	$HCl(g)\rightarrow$	$NH_4Cl(s)$
Ini moles	$\frac{5\times 3}{RT}$	$\frac{2\times 4}{RT}$	–
Final moles	$\frac{7}{RT}$	–	$\frac{8}{RT}$

$$P_fV_f = n_fRT$$

$$\Rightarrow P_f\times 7 = \frac{7}{RT}\times RT$$

$$\Rightarrow P_f = 1\,\text{atm}$$

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43.

$$P_{\text{right}} = \frac{80 \times 25}{20} = 100\,\text{cm of } Hg$$

$$P_{\text{left}} = 100 + 10 = 110\,\text{cm of } Hg$$

$$\alpha = \left[\frac{(PV)_{final}}{(PV)_{inital}} - 1\right] \text{for } NH_3$$

$$= \frac{110 \times 30}{80 \times 25} - 1 = 0.65$$

44. After addition of $0.2\,M\ NaOH$, since $pH = pK_a$, a buffer solution consisting of equal moles of $HOCl$ and $NaOCl$ has beed formed.

$$\left(pH = pK_a \frac{[NaOCl]}{[HOCl]}\right)$$

Let volume of original $HOCl$ solution = $V\,mL$

$HOCl + NaOH \rightarrow NaOCl + H_2O$

1 mol of $0.2V\ (20 \times 0.2) = 4$ -

Equilibrium $(0.2V - 4)$ - 4

$0.2V - 4 = 4$

$\Rightarrow V = 40\,ml$

After addition of $40\,ml\ NaOH$

$HOCl + NaOH \rightarrow NaOCl + H_2O$

mmol 0.2×40 0.2×40

equilibrium - - 8

$$\therefore [NaOCl] = \frac{8}{40 + 40} = 0.1M$$

$$pH = \frac{1}{2}(pK_w + pK_a + \log C)$$

$$= \frac{1}{2}(14 + 7.5 - 1) = 10.25$$

45.

	Ka	$\Delta G^\circ C\,(\text{kJ/mole})$	$\Delta H^\circ C\,(\text{kJ/mole})$	$T\Delta S^\circ\,(\text{kJ/mole})$
$HCOOH$	17.7×10^{-5}	+21	–0.4	–21.3kJ
CH_3COOH	1.75×10^{-5}	+27.2	–0.5	–27.6kJ

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CH_2-COOH $\vert$ Cl	136×10^{-5}	+16	–4.6	–21kJ

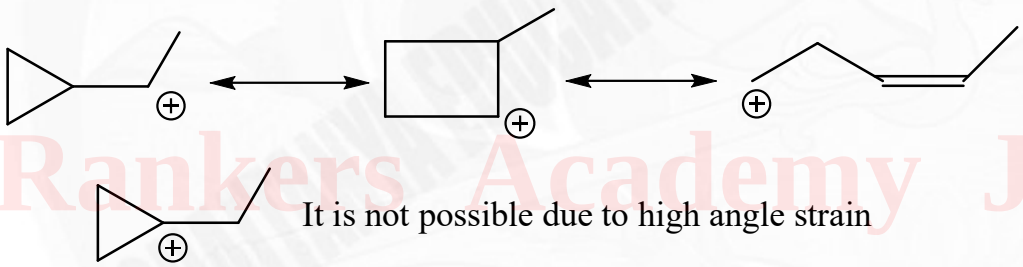
$\rightarrow$ Ions formed in ionization are highly solvated, as a result ΔS° is $-ve$

$\rightarrow$ From the above data, greater contribution for acidic strength is ΔS° (i.e solvation of ions only)

$\rightarrow$ $E.D.G$ increases the charge density at $-COO^-$ as a result, solvation of $-COO^-$ increases. Then ΔS° become more $-ve, \Delta G^\circ$ becomes more $+ve$. Finally acidic strength of carboxylic acid decreases.

$\rightarrow$ E.W.G decreases the charge density at $-COO^-$ a sa result , salvation of $-COO^-$ decreases. Then ΔS° v=become less $-ve, \Delta G^\circ$ becomes less $+ve$. Finally acidic strength of carboxylic acid increases.

46.



It is not possible due to high angle strain

47. NO_2^- – bond order 1.5

NO_3^- – bond order 1.33

$N_2O < N_2O_4 < N_2O_3$ (Order of N - N bond length)
113 175 186 (*in pico meters*)

48. Carbon shows more tendency to form $p\pi - p\pi$ bonds where as silicon produces unstable $3p\pi - 3p\pi$ bonds

49. Since partial pressure of O_2 is constant, upon increasing volume, equilibrium shift in the forward direction

50. a) $\Delta V_{1-2} = O$ (siothermal process)

$\Delta V_{2-3} = W_{2-3}$ (adiabatic process)

b) $\Delta S_{1-2} + \Delta S_{2-3} = \Delta S_{1-3}$ ($\because$ 'S' is a state function)

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But $\Delta S_{2-3} = 0$ (Reversible adiabatic process)

$\therefore \Delta S_{1-2} = \Delta S_{1-3}$

C) W is not a state function

D) $Q_{1-3} + W_{1-3} = \Delta U_{1-3} = \Delta U_{2-3} = W_{2-3}$

51. Flourination: Step – I is exothermic, Step – II is exothermic, Overall reaction is highly exothermic.

Chlorination: Step – I is slightly exothermic, Step – II is exothermic, overall reaction is exothermic.

Bromination: Step – I is endothermic, Step – II is exothermic, overall reaction is exothermic.

Iodination: Step – I is endothermic, Step – II is exothermic, overall reaction is endothermic.

Step – I is R.D.S, Step – II is fast

52. Boiling point (K) - $B > Al > Ga > In > Tl$

Ionic radius - $Sb^{3+} < Bi^{3+} < N^{3-} < P^{3-} < As^{3-}$

Atomic radius (pm) - $Tl > In > Al > Ga > B$

Density $\left(g / cm^3\right)$ at 298 K - $N < P < As < Sb < Bi < Pb$

53. $3d_{xy} \rightarrow 2$ nodal planes intersecting in a straight line

$4d_{x^2-y^2} \rightarrow 2$ nodal planes and a spherical node $\rightarrow$ 1 straight line, 2 circles

$3d_{z^2} \rightarrow 2$ concial nodal surfaces $\rightarrow$ Point

$4d_{z^2} \rightarrow 2$ concial nodal surfaces and a spherical node $\rightarrow$ Point, 2 circles

54. $1,2 \rightarrow$ Polytropic process. $3 \rightarrow$ Reversible adiabatic process, $4 \rightarrow$ isothermal

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